CHARLES DARWIN UNIVERSITY
College of Engineering, Information Technology and Environment

PRT565 – Machine Learning and Artificial Intelligence

Assessment 4: Group Report

Detailed Reporting on Assessment 3 (Group Presentation)

Breast Cancer Diagnosis Prediction

A Comparative Study of Machine Learning and Deep Learning Classification Models

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Master of Information Technology (Cyber Security)
Campus: Sydney
Submission date: October 2026
Assessment 3 recorded presentation: https://youtu.be/K0MlCXWd6a4

Group Information and Member Contributions

Project title: Breast Cancer Diagnosis Prediction – A Comparative Study of Machine Learning and Deep Learning Classification Models

Group: SYD-G1-22
Unit: PRT565
Dataset: Breast Cancer Wisconsin (Diagnostic), UCI Machine Learning Repository

This report is the detailed written account of the project our group presented in Assessment 3 (recorded presentation: https://youtu.be/K0MlCXWd6a4). It expands on every stage shown in the presentation, from the problem and dataset through to the algorithms, evaluation measures and results, and adds the full exploratory analysis and the reasoning behind each decision.
The report was divided into three parts through a group discussion. Each member wrote the part allotted to them, and every member reviewed the other two parts for accuracy and consistency before submission.

Member	Name and ID	Parts completed
Member 1	Jessica Karki (S398272)	Sections 1–3: Problem Description, Dataset Description, Dataset Analysis and Preprocessing
Member 2	Diwash Chettri (S398690)	Section 4: Choice and Description of the Algorithms and Key Implementation Steps
Member 3	Binu Regmi (S392323)	Sections 5–7: Overview of Model Evaluation Measures, Results Obtained and Overall Understanding
		Total

Table of Contents

Group Information and Member Contributions *2*
1. Problem Description and Motivation *6*
1.1 Background 6
1.2 Problem Statement 6
1.3 Motivation and Aims 7
2. Dataset Description *7*
2.1 Source 7
2.2 Features 7
2.3 Target Variable and Class Balance 9
3. Dataset Analysis, Feature Selection, Transformation, and Engineering *9*
3.1 Loading and Data-Quality Checks 9
3.2 Feature Distributions by Class 10
3.3 Comparing All Mean Features Across Classes 11
3.4 Correlation and Multicollinearity 12
3.5 Multi-Variable Relationships 13
3.6 A Lower-Dimensional View with PCA 14
3.7 Feature Selection Decision 14
3.8 Train/Test Split and Feature Scaling 15
4. Choice and Description of the Algorithms *17*
4.1 Why These Five Algorithms 17
4.2 Decision Tree 17
4.3 Random Forest 18
4.4 Gaussian Naive Bayes 18
4.5 Support Vector Machine (RBF Kernel) 19
4.6 Artificial Neural Network (Multilayer Perceptron) 19
4.7 How the Hyperparameters Were Chosen 20
4.8 Key Implementation Steps 20
5. Overview of Model Evaluation Measures *22*
5.1 The Confusion Matrix 22
5.2 Accuracy 22
5.3 Precision 22
5.4 Recall (Sensitivity) 22
5.5 Supporting Measures: F1-Score and ROC-AUC 23
6. Results Obtained *23*
6.1 Overall Performance 23

6.2 Confusion Matrices and the Malignant Class 24
6.3 ROC Curves 25
6.4 Feature Importance 26
7. Overall Understanding and Conclusion *26*
7.1 What the Results Tell Us 26
7.2 Limitations 27
7.3 Future Work 27
7.4 Conclusion 28
References *29*

Table of Figures:

Figure 1: Class distribution of the diagnosis label 9
Figure 2: Code: loading the dataset and checking data quality 10
Figure 3: Distributions of four features, split by diagnosis 10
Figure 4: Standardized boxplots of the ten mean features by diagnosis 11
Figure 5: Pearson correlation matrix of the 30 features 12
Figure 6: Pair plot of four “worst” features coloured by diagnosis 13
Figure 7: Principal Component Analysis: 2-D projection (left) and explained variance (right) 14
Figure 8:Code: stratified split and feature scaling 15
Figure 9: Six features before scaling (log axis) and after standardization 15
Figure 10: The top three levels of the trained Decision Tree (thresholds are on standardized values) 18
Figure 11:Code: defining the five models and their hyperparameters 20
Figure 12:Code: common training and evaluation loop 21
Figure 13: Comparison of all five metrics across the models (y-axis starts at 0.88) 23
Figure 14: Confusion matrices on the test set (rows = actual, columns = predicted) 24
Figure 15: ROC curves for the five models on the test set 25
Figure 16:Code: extracting feature importance from the Random Forest 26
Figure 17: Random Forest: top 15 features by mean decrease in impurity 26

Table 1 : The ten nuclear characteristics measured in each image 8
Table 2 : Summary of the preprocessing pipeline 16
Table 3: Summary of the five models 21
Table 4: Test-set results (precision, recall, and F1 for the benign class, scikit-learn default) 23
Table 5: Errors and metrics with malignant as the positive class (42 malignant test cases) .. 24

Part 1 – Member 1: Jessica Karki (S398272)

Problem Description, Dataset Description, Dataset Analysis and Preprocessing

1. Problem Description and Motivation

1.1 Background

Globally, breast cancer has become the most common form of cancer among women and one of the world's top causes of cancer-related deaths. According to recent global estimations, in 2022, there will be approximately 2.3 million newly diagnosed cases of breast cancer and approximately 670,000 related deaths (Bray et al., 2024). The ability to survive breast cancer depends greatly upon the timeliness and accuracy in which a person's tumour is diagnosed. If a malignant tumour is diagnosed at an early stage, there exist many more treatment options available that are less invasive and much more effective than if the same tumour were diagnosed at a later date.

One method of diagnosis is the Fine Needle Aspirate (FNA) Biopsy. During an FNA Biopsy, a very thin needle is inserted into the area where a woman's breast lump exists. A small number of cells are extracted from the lump via the needle and placed onto a glass slide. The cells are then stained with various dyes. Afterward, the digitized image of the cell nuclei is viewed by a pathologist. Based upon the appearance of the nuclei, i.e., their size, shape and texture, the pathologist determines if the cells appear to be either benign (non-cancerous) or malignant (cancerous). Due to these varying appearances of the nuclei and the subjective nature of this assessment process, it is possible for two different pathologists to assess the same sample of cells differently, particularly in situations where cells exhibit features that fall somewhere between clearly benign and clearly malignant, or when a high volume of samples need to be evaluated (Street et al., 1993).

1.2 Problem Statement

The project studied the problem of binary classification. We used 30 numerical measurements from a digitized fine needle aspiration (FNA) image of a breast mass to determine if the tumour was malignant or benign. Each model produced a class label and a corresponding probability. The quality of a model is determined by how accurately it predicted the confirmed diagnosis for cases it did not see during training.

Rather than developing one model, we compared five different models representing the major types of supervised learning; a decision tree, a random forest ensemble, a Gaussian Naive Bayes probabilistic model, a support vector machine with a radial basis function (RBF) kernel, and an artificial neural network in the form of a multilayer perceptron (MLP). Comparing traditional machine learning approaches to deep learning on the same dataset allowed us to evaluate which method provides the most accurate diagnostic assistance for this type of structured medical data.

1.3 Motivation and Aims

Our objective is to develop a working model that can provide a fast, reliable estimate of whether a given sample is likely to contain cancer cells. As such, the resulting model would provide an additional "second opinion" to clinicians, allow them to prioritize urgent cases, and minimize the risk of misdiagnosis that could lead to delayed treatment. The two main types of errors in this context are not equally important. A false negative (a malignant tumour classified as benign) may cause significant delays in treatment and possibly put someone's life at risk, whereas a false positive (a benign tumour classified as malignant) will typically only result in increased anxiety and/or additional tests. We chose to evaluate precision, recall, F1-score, and ROC-AUC in addition to accuracy because these metrics better reflect the consequences of both error types.

The objectives of the project were:

- to analyse the Breast Cancer Wisconsin (Diagnostic) dataset using summary statistics and visualization techniques;
- to correctly preprocess the data so that all models are trained and tested using identical conditions;
- to train and evaluate five classification models representing different families of models; and
- to determine which model produced the highest performance and what features of the tumour were responsible for driving the model's predictions.

2. Dataset Description

2.1 Source

The Breast Cancer Wisconsin (Diagnostic) Dataset was developed by Dr. William Wolberg, Olvi Mangasarian, and W. Nick Street at the University of Wisconsin. This dataset was later made available to the public via the UCI Machine Learning Repository (Wolberg et al., 1995). The exact same dataset is also available in Scikit-Learn as load_breast_cancer(). That's how we loaded it into Python. You will find a CSV version of this dataset (breast_cancer_wisconsin.csv) in our submission so that you may view the raw data.

2.2 Features

The dataset contains **569 samples** (one per patient) and **30 numeric input features**. Each sample was produced by measuring every cell nucleus visible in the FNA image. Ten characteristics were recorded for each nucleus:

Table 1 : The ten nuclear characteristics measured in each image

Characteristic	**What it describes**
Radius	Mean distance from the center of the nucleus to points on its boundary
Texture	Standard deviation of grey-scale values (variation in staining)
Perimeter	Length of the nucleus boundary
Area	Area enclosed by the boundary
Smoothness	Local variation in radius lengths
Compactness	Perimeter² / area − 1.0
Concavity	Severity of concave (inward-curving) portions of the contour
Concave points	Number of concave portions of the contour
Symmetry	Symmetry of the nucleus shape
Fractal dimension	"Coastline approximation" − 1 (boundary complexity)

For each characteristic, three summary statistics were calculated over all nuclei in the image: the **mean**, the **standard error (SE)**, and the **"worst"** value (the mean of the three largest values). This gives 10 × 3 = 30 features, named, for example, *mean radius*, *radius error,* and *worst radius*. All features are continuous real numbers, but they are on very different scales: *mean area* ranges from 143.5 to 2,501, while *mean smoothness* only ranges from 0.053 to 0.163.

2.3 Target Variable and Class Balance

The objective of this study was to diagnose. For scikit-learn, the objective was coded as 0=Malignant and 1=Benign. When you calculate precision and recall for your dataset, scikit-learn will treat label 1 as the default "positive" class.

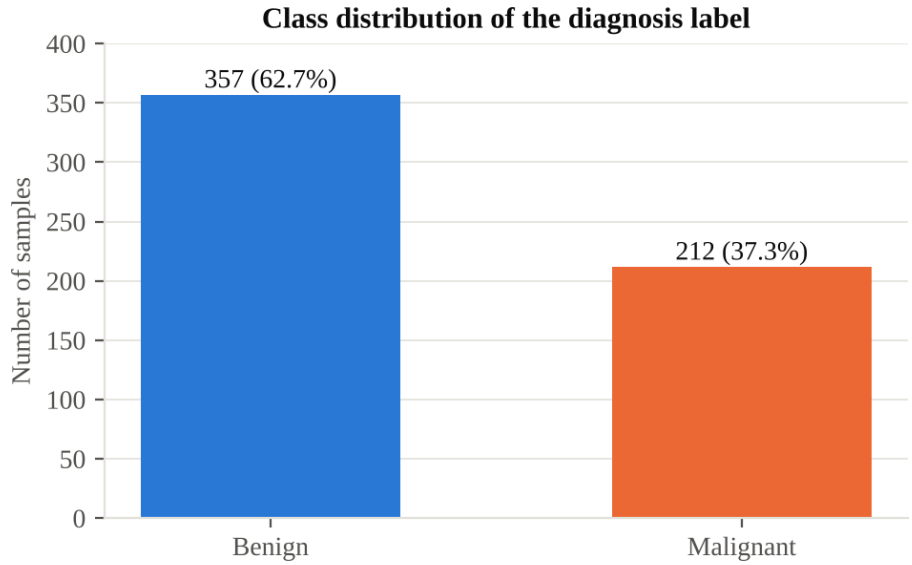


Figure 1: Class distribution of the diagnosis label

What we've learned: The data set contains 357 Benign (62.7%) and 212 Malignant (37.3%) examples.

Therefore, while there is a moderate imbalance in the data, it is not severe. There were sufficient examples of the minority class (over 200) for further analysis. A model that predicts "benign" for every example will have an accuracy of 62.7%; therefore, the accuracy of any model must be significantly higher than 62.7% before it can be considered useful. A random split of the training and testing sets could alter the class ratio; therefore, we used a stratified split (Section 3.8).

3. Dataset Analysis, Feature Selection, Transformation, and Engineering

3.1 Loading and Data-Quality Checks

The first step was to load the data into a pandas DataFrame and confirm its shape and quality. The code below checks the dimensions, missing values, duplicate rows, and the class counts.

```
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")   # 0 = malignant, 1 = benign

print("Shape:", X.shape)                                    # (569, 30)
print("Missing values:", int(X.isnull().sum().sum()))       # 0
print("Duplicate rows:", int(X.duplicated().sum()))         # 0
print(y.map({0: "Malignant", 1: "Benign"}).value_counts())
```

Figure 2: Code: loading the dataset and checking data quality

We verified that the final data set consisted of a 569 x 30 feature matrix with no missing values and no duplicate rows. As such, no additional imputation or row removal was necessary. Since the features were generated automatically using image analysis software rather than entered manually, this was to be expected.

3.2 Feature Distributions by Class

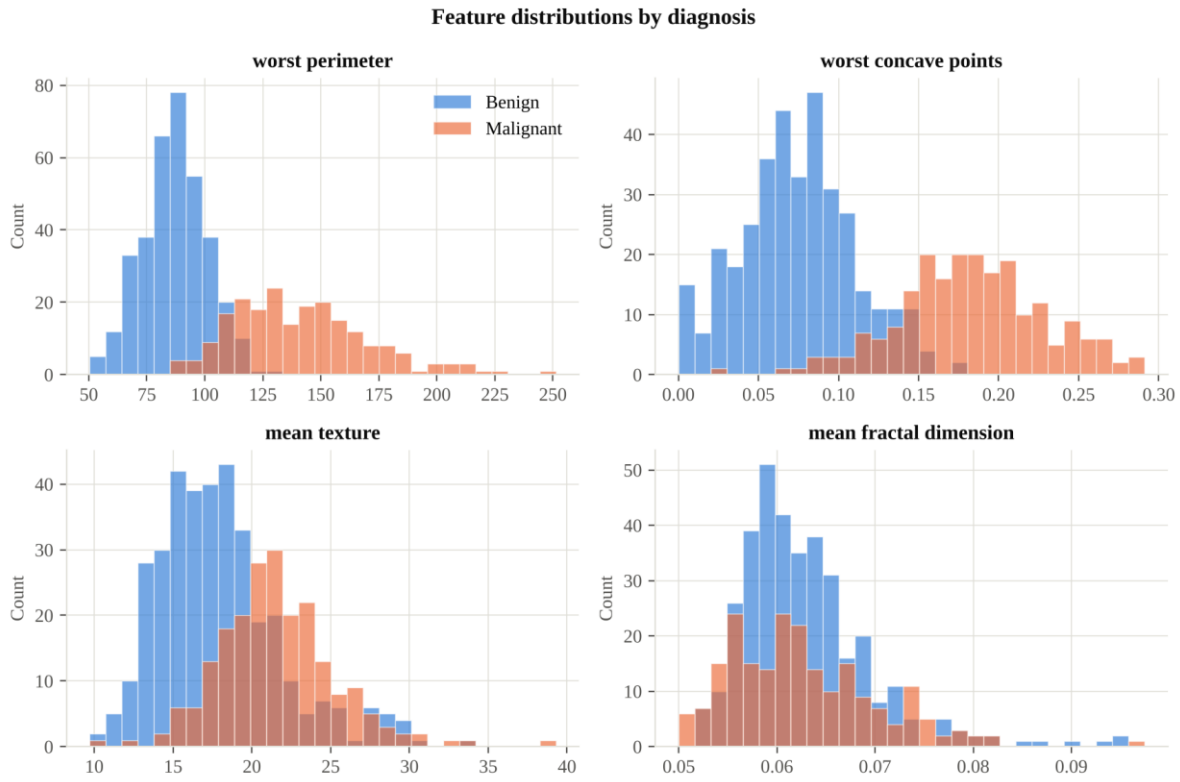


Figure 3: Distributions of four features, split by diagnosis

What we observed: There was a significant difference in the effectiveness of each variable as shown by the four histograms. The differences between the benign and malignant distributions for Worst Perimeter and Worst Concave Points were minimal; the benign distributions clustered around lower perimeters (approximately 60-110) and fewer concave points than the malignant distributions which were spread to the right.

Therefore, if one were to set a single threshold on either of these variables, he/she could distinguish between the majority of cases. Although Mean Texture showed a clear separation (malignant cases were more textured), there was still considerable overlap.

3.3 Comparing All Mean Features Across Classes

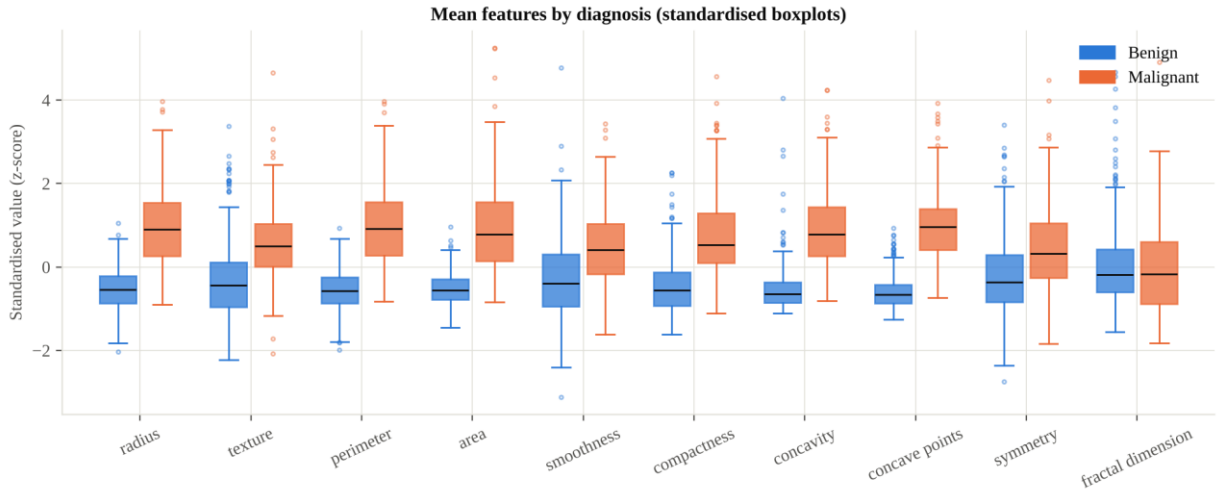


Figure 4: Standardized boxplots of the ten mean features by diagnosis

We found that if you standardize each of the 10 mean features, the comparison between the two groups (malignant vs. benign) will be fair. The malignant box is positioned above the benign box for radius, perimeter, area, concavity, and concave points. This indicates that malignant cells have larger and more irregularly shaped nuclei than benign cells. Texture, compactness, smoothness, and symmetry exhibit smaller shifts. Fractal dimension does not exhibit any shift.

In addition, there were many outlier values. Nearly all of these outliers occurred at the higher end of the range. The outlier values represent actual biological measurements of large, irregular tumours and not entry error.

Therefore, we chose to retain the outlier values because if we had removed them, we would have lost those examples of malignancy that were the most obvious examples.

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3.4 Correlation and Multicollinearity

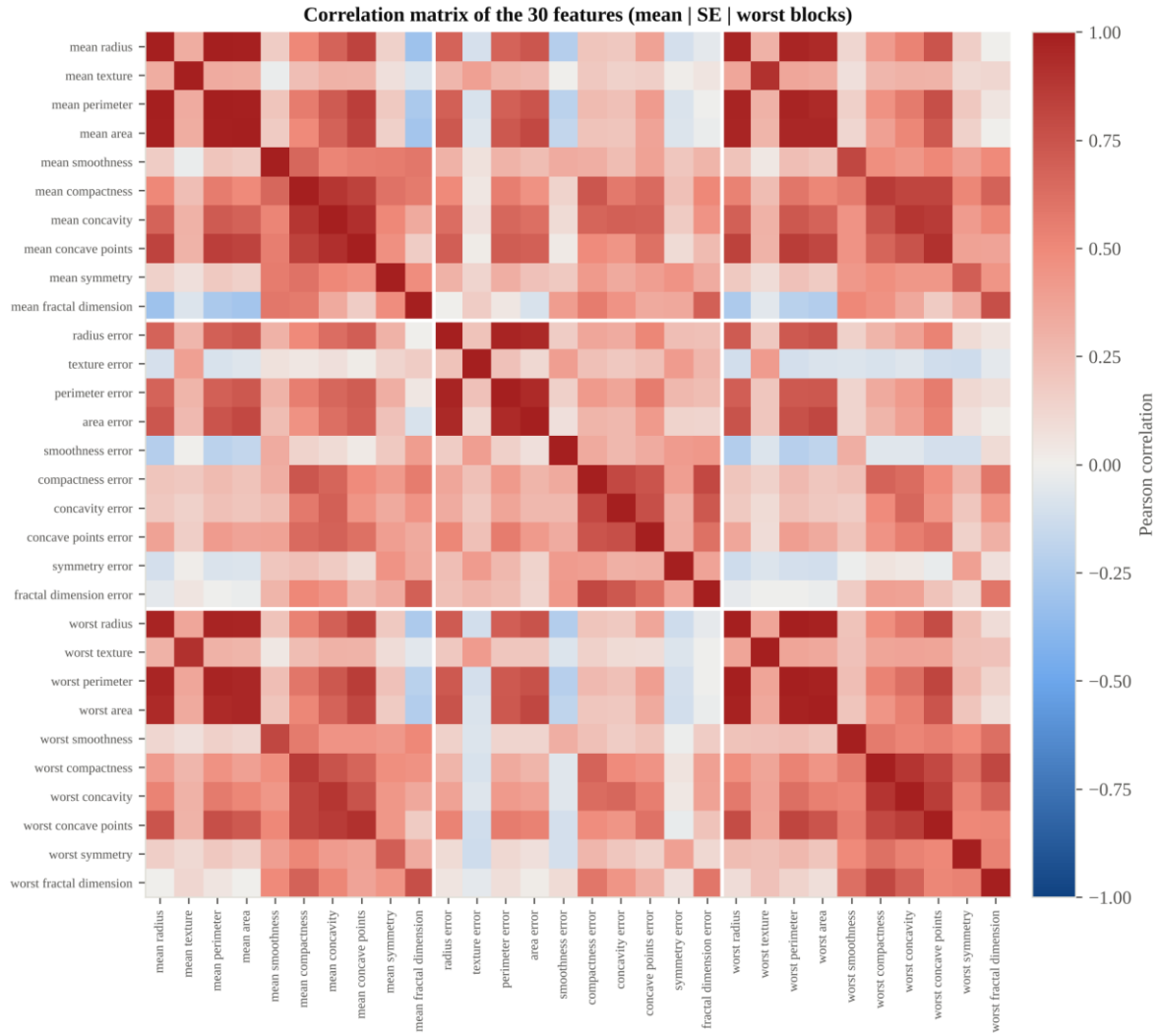


Figure 5: Pearson correlation matrix of the 30 features

What we found out: The heat map shows a very high degree of multicollinearity among many variables.

For example, the mean radius and the mean perimeter are almost perfectly correlated with each other (r = 0.998). This is not surprising, since both the perimeter and the area depend on the radius.

In addition, there were 21 pairs of variables that had absolute values greater than 0.9.

Another set of correlated variables includes compactness, concavity, and concave points; these three variables measure how irregular the tumour's shape is. Also, each "mean" version of the features was highly correlated with its corresponding "worst" version. This relationship can be seen by the bright diagonal lines in the off-diagonal blocks.

Overall, the SE features were less correlated with the other features than were the mean or worst versions. With regard to correlation with the diagnosis of malignancy, the highest individual correlations are worst concave points (r = 0.79), worst perimeter (0.78), mean concave points (0.78), and worst radius (0.78).

On the other hand, some of the lower correlated variables include mean fractal dimension, texture error, and symmetry error.
190 words

3.5 Multi-Variable Relationships

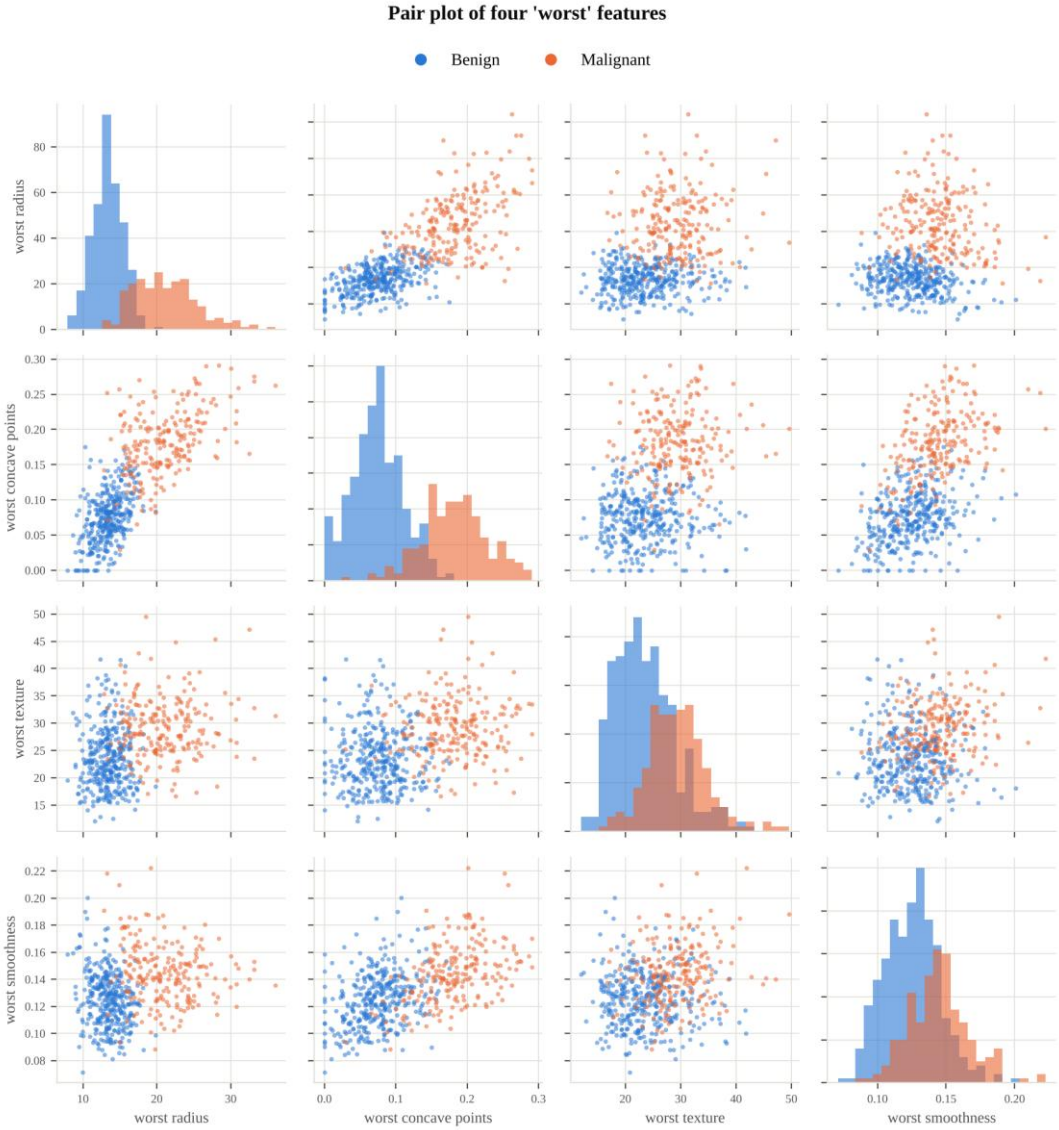


Figure 6: Pair plot of four “worst” features coloured by diagnosis

What we learned: The pair plot demonstrates how different combinations of variables can help differentiate between the two groups. The strongest separation occurs when comparing

the worst radius versus the worst concave points. In this case, the two groups are separated into two distinct clouds of data points; there is very little overlap at the boundary of these clouds, and the boundary itself is curved rather than straight (i.e., neither vertically nor horizontally).

As a result, I would expect a model that uses both a "size" variable (worst radius) and a "shape" variable (worst concave points) to be more effective than one that uses either variable individually. Also, since the boundary is not linear, I would also expect a model capable of creating non-linear boundaries (such as an RBF kernel support vector machine) to have a relatively high level of accuracy.

3.6 A Lower-Dimensional View with PCA

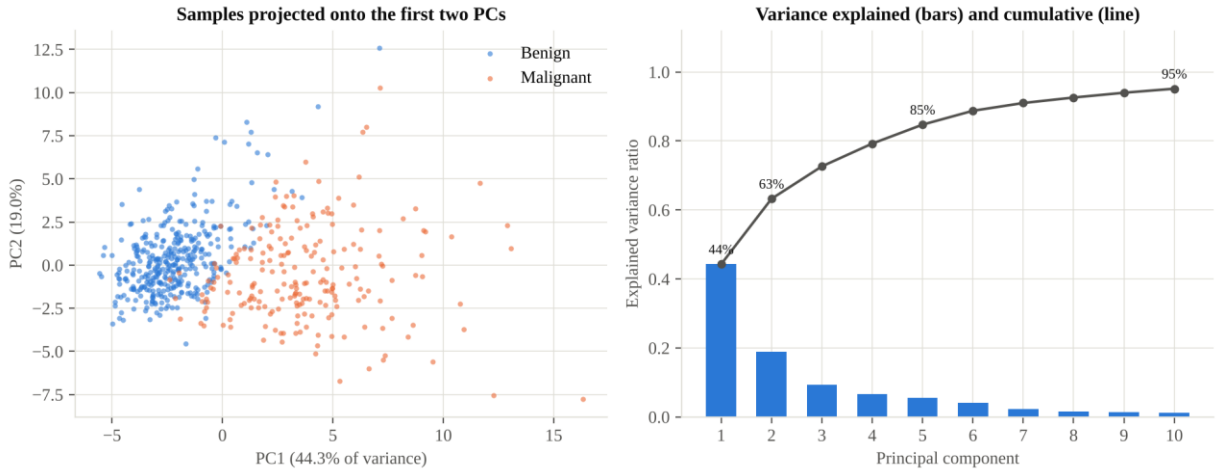


Figure 7: Principal Component Analysis: 2-D projection (left) and explained variance (right)

What we learned: Only Principal Component Analysis (PCA) was performed for exploratory purposes. The first PCA accounts for 44.3% of the total variability; the first two PCAs account for 63.2%; the fifth PCA accounts for 84.7%; and the tenth PCA accounts for 95.2%.

Therefore, the 30 features contained a large amount of redundant information. Furthermore, in a 2-D visualization, the benign samples formed a tight cluster on the left side of the chart, while the malignant samples were spread out on the right. Therefore, the two classes appeared to be almost linearly separable even when using only two dimensions.

3.7 Feature Selection Decision

We chose not to eliminate the highly correlated features or replace them with Principal Components for several reasons. First, the number of features (30) relative to the number of training observations (569) makes this a very low-dimensional data set and thus does not pose a computational problem.

Secondly, both models (Tree-Based Models and Support Vector Machines) used in this study can handle highly correlated features.

For example, Tree-Based Models will only select one of the highly correlated features if there is no difference in their predictive power. Support Vector Machines and Multi-Layer Perceptron's will create a weighted combination of highly correlated features based on how much each feature contributes to the prediction.

Lastly, we used the named features directly to determine feature importance with the Random Forest classifier after training (see Section 6.4).

Therefore, our only use of feature engineering was to scale the features since the original authors had previously engineered the 30 features from the raw image data.

3.8 Train/Test Split and Feature Scaling

```
# Stratified 80/20 split keeps the 63:37 benign:malignant ratio
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)

# Fit the scaler on the training data only (no test-set leakage)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s  = scaler.transform(X_test)
```

Figure 8:Code: stratified split and feature scaling

The data was split into **80% training (455 samples)** and **20% testing (114 samples)** using *stratify=y* and *random_state=42*. Stratification kept the class ratio identical in both sets (285 benign / 170 malignant in training and 72 benign / 42 malignant in testing), and the fixed random state makes the experiment reproducible.

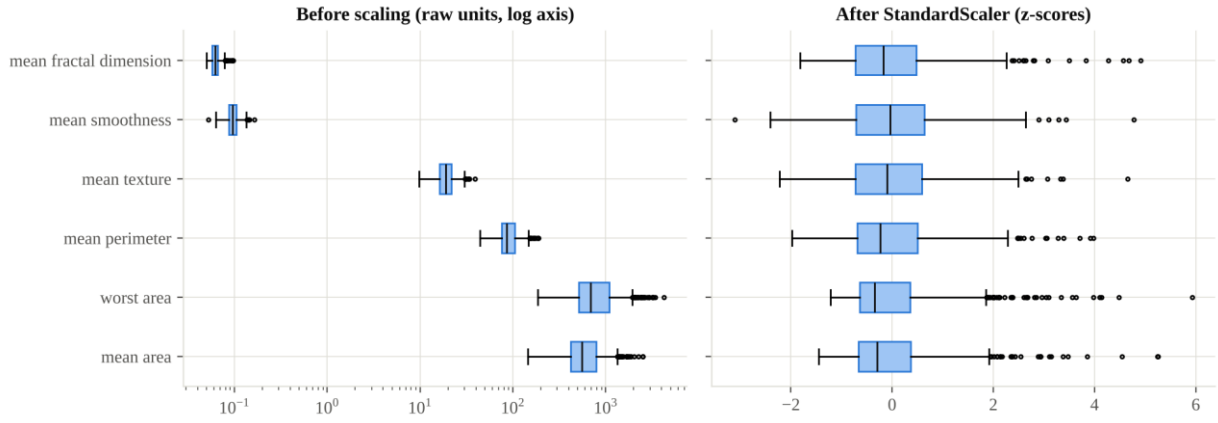


Figure 9: Six features before scaling (log axis) and after standardization

What we learned: Before scaling, the features span more than four orders of magnitude (left panel, log axis). Distance-based and gradient-based models would let large-valued features, such as area, dominate small ones like smoothness, purely because of their units. After **StandardAero**, each feature has a mean of 0 and a standard deviation of 1 (right panel), so all features start on an equal footing. The scaler was fitted to the training set only and then applied to the test set, thereby avoiding leaking test information into training. Scaled data was used for the Decision Tree, Random Forest, SVM, and ANN. Gaussian Naive Bayes used the original unscaled values because it models each feature with its own mean and variance and is therefore unaffected by scale.

Table 2 : Summary of the preprocessing pipeline

Step	Action	Reason
1	Load 569 × 30 data; check missing values and duplicates	Confirm data quality (none found)
2	Exploratory analysis (Figures 3–7)	Understand separability, outliers and redundancy
3	Keep all 30 features and outliers	Small dataset; outliers are genuine malignant cases
4	Stratified 80/20 split, random_state = 42	Fair, reproducible evaluation with the same class ratio
5	StandardScaler fitted on training data	Equal feature scales; no test-set leakage

Part 2 – Member 2: Diwash Chettri (S398690)

Choice and Description of the Algorithms and Key Implementation Steps • 1,302 words

4. Choice and Description of the Algorithms

4.1 Why These Five Algorithms

As such, the exploratory data analysis has demonstrated that the two classes can be easily distinguished by their size and shape features, and that the decision boundary separating them is curvilinear rather than linear.

Additionally, the features are highly correlated.

Therefore, we sought out a variety of classification techniques with differing underlying assumptions about the data, allowing us to compare them and understand how they perform relative to one another, and ultimately to provide insight into what type of classification problem this really represents. From each family of five different types of models, we chose one model: a decision tree as a rule-based model; a random forest as an ensemble model; a Gaussian Naive Bayes as a probabilistic model; an SVM using an RBF kernel as a kernel method; and a multi-layer perceptron as a neural network model. We implemented all five models with Scikit-learn.

153 words

4.2 Decision Tree

Decision Trees classify an observation by answering a series of binary (yes/no) questions based on the values of the predictors or variables associated with the observation. An example question would be "Is the worst radius <= 0.5?" The training algorithm begins with all observations at the root node and then iteratively selects the variable and corresponding value that most efficiently separates the classes. The selection criterion is the reduction in Gini impurity (Breiman et al., 1984), defined as Gini = 1 - sum over i of p(i)^2, where p(i) denotes the percentage of class i in a particular node. A "pure" node will have a Gini value of zero since there are no other classes represented in the node. Splitting each node into two child nodes allows the process to repeat until a stopping condition occurs. Since a decision tree can learn every possible combination of predictor variables, it can memorize the training data. We limited it to **max_depth = 5**, which produced 15 leaf nodes.

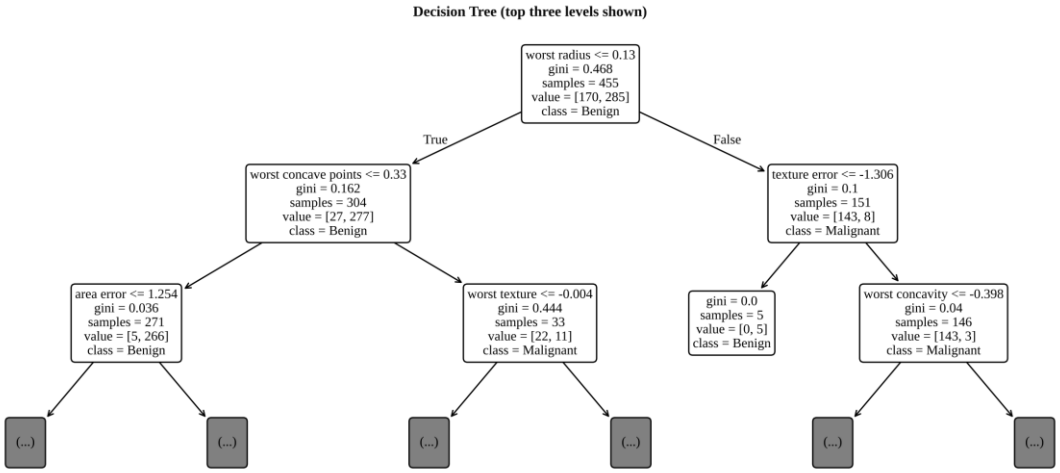


Figure 10: The top three levels of the trained Decision Tree (thresholds are on standardized values)

Figure 10 demonstrates how interpretable this model is. The root split uses the worst radius. Samples with values below the threshold (304 training samples) are 91% benign, and those with values above the threshold (151 training samples) are 95% malignant. The second split uses the worst-concave points and texture errors to clean up the remaining mixed cases. This is consistent with what we found during our exploratory analysis; size and concavity provided the best separation. A major drawback of using a single tree is its very high variance. Changes to the training data may cause the splits to vary greatly. Also, since all the splits are axis-parallel, they form a "staircase" approximation to the diagonal line shown in Figure 6.

4.3 Random Forest

The Variance Problem is addressed in Random Forests (Breiman, 2001) by training multiple decision trees independently and then combining them via voting. Each of our 200 trees was trained on a bootstrap sample (a random sample drawn with replacement) of the training data. At each node, the tree could only evaluate a subset of randomly selected features (by default $\sqrt{30} \approx 5$ features). Because there are two sources of randomness in how each tree is built – i.e., both the training set and the available features – they will be somewhat different from one another.

Therefore, the error made by one tree will likely be cancelled out by the error made by another tree when their outputs are averaged together. The predicted probability for a given instance from the Random Forest model is simply the average of the probabilities generated by each individual tree.

4.4 Gaussian Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem:
$P(class \mid x) = P(class) \times \Pi\, P(x(j) \mid class) / P(x)$

It is called “naive” because it assumes every feature is independent of the others once the class is known. The Gaussian version models each feature within each class with a normal distribution, parameterized by the class mean and variance. Prediction then picks the class with the highest posterior probability. Training is extremely fast, as it only requires computing means and variances. We included it as a **baseline**. Our correlation analysis showed that the independence assumption is clearly violated here (for example, radius, perimeter, and area are almost perfectly correlated), so the model effectively counts the same evidence several times. Naive Bayes often still classifies well in this situation, because only the ranking of probabilities matters, not their exact values (Zhang, 2004). Because each feature is modelled with its own mean and variance, scaling does not change the results, so the model was trained on the unscaled data.

4.5 Support Vector Machine (RBF Kernel)

A Support Vector Machine (Cortes & Vapnik, 1995) looks for the decision boundary that separates the classes with the **widest possible margin**. Only the training points closest to the boundary, called **support vectors**, determine its position. To handle boundaries that are not straight lines, the SVM uses a kernel function that implicitly maps the data into a higher-dimensional space where a linear separation is possible. We used the **radial basis function (RBF)** kernel:

$K(x, x') = exp(-\gamma \, ||x - x'||^2)$

which measures similarity based on distance. Because the kernel is distance-based, feature scaling is essential, so the SVM was trained on standardized data. We used the scikit-learn defaults of **C = 1.0** (the penalty for misclassified training points, balancing margin width against training errors) and **gamma = “scale”** (which equals 1/30 on standardized data). The trained model used 97 of the 455 training samples (21%) as support vectors. Setting *probability=True* makes the SVM produce class probabilities through Platt scaling, which we needed for the ROC curve.

4.6 Artificial Neural Network (Multilayer Perceptron)

To represent deep learning, we built a feed-forward **Multilayer Perceptron** (LeCun et al., 2015). It has an input layer of 30 neurons (one per feature), **two hidden layers with 64 and 32 neurons**, and a single output neuron with a logistic (sigmoid) activation that outputs the probability of the benign class. Each hidden neuron computes a weighted sum of its inputs and passes it through the **ReLU** activation function, max(0, z), which allows the network to learn non-linear relationships. The weights are learned via backpropagation (Rumelhart et al., 1986) with the Adam optimizer (Kingma & Ba, 2015) to minimize log-loss.

Neural networks can easily overfit small datasets; our network has 4,097 trainable weights, but only about 410 training samples. To control this, we enabled **early stopping**: 10% of the training data is held out as a validation set, and training stops when the validation score has not improved for 10 consecutive epochs. Training stopped after 25 epochs with a best validation accuracy of 95.65%. A small L2 penalty (alpha = 0.0001) was also applied by default.

4.7 How the Hyperparameters Were Chosen

Because the goal of this study was to fairly compare the models rather than optimize one model's performance, we intentionally selected parameters for each model that were simple and well-known. These are also known as "default" or "standard" settings. We chose these settings instead of spending time searching for optimal parameters. Each decision made by us was associated with a particular risk. The Decision Tree was set to a depth of 5. This allows the Decision Tree to use three different features (the size feature, the shape feature, and the texture feature (the pattern shown in Figure 10)) to make predictions.

However, the depth of 5 does not allow the Decision Tree to grow so large that there would be very few training instances in each leaf node. As such, the Decision Tree will not over-fit. The Random Forest was configured to have 200 trees. At this point, averaging predictions across the trees will not change significantly.

Furthermore, training the Random Forest was completed in less than 1 second. The SVM was configured to utilize the default C and gamma settings. The default settings were selected because they have been demonstrated to perform well on standardized data, and our preprocessing technique resulted in standardized data. The ANN was configured to include two hidden layers with decreasing numbers of nodes (64 in the first layer, 32 in the second). This configuration provided the ANN with sufficient capacity to develop non-linear relationships between multiple features. The early stopping technique was utilized as an automatic mechanism to prevent the ANN from memorizing the training data. There are virtually no adjustable parameters for Naive Bayes.

Therefore, Naive Bayes serves as an excellent baseline. Using identical, lightly tuned settings across all five models allows us to attribute differences observed in the experimental results to how well each model's assumptions align with the data's characteristics, rather than to the time spent optimizing each model.

4.8 Key Implementation Steps

All five models were defined in a single dictionary so they could be trained and evaluated using exactly the same loop. A fixed *random_state = 42* was used wherever randomness is involved, so the results can be reproduced exactly.

```
models = {
    "Decision Tree": DecisionTreeClassifier(max_depth=5, random_state=42),
    "Random Forest": RandomForestClassifier(n_estimators=200, random_state=42),
    "Naive Bayes":   GaussianNB(),
    "SVM (RBF)":     SVC(kernel="rbf", probability=True, random_state=42),
    "ANN (MLP)":     MLPClassifier(hidden_layer_sizes=(64, 32), activation="relu",
                                   solver="adam", early_stopping=True,
                                   max_iter=1000, random_state=42),
}
uses_unscaled = {"Naive Bayes"}   # Gaussian NB works on the raw features
```

Figure 11:Code: defining the five models and their hyperparameters

The training loop selects the scaled or unscaled version of the data for each model, fits the model to the training set, predicts labels and probabilities on the test set, and records all metrics. Using one loop for each model ensures that no model gains an unfair advantage.

```
for name, model in models.items():
    Xtr, Xte = (X_train, X_test) if name in uses_unscaled else (X_train_s, X_test_s)
    model.fit(Xtr, y_train)
    y_pred = model.predict(Xte)
    y_prob = model.predict_proba(Xte)[:, 1]
    rows.append({
        "Model":     name,
        "Accuracy":  accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall":    recall_score(y_test, y_pred),
        "F1-Score":  f1_score(y_test, y_pred),
        "ROC-AUC":   roc_auc_score(y_test, y_prob),
    })
    print(confusion_matrix(y_test, y_pred))
```

Figure 12:Code: common training and evaluation loop

Table 3: Summary of the five models

Model	**Key settings**	**Input**	**Type of decision boundary**
Decision Tree	Gini, max_depth = 5	Scaled	Axis-parallel, step-shaped
Random Forest	200 trees, max_features = √n, bootstrap	Scaled	Averaged steps (smoother)
Gaussian NB	Default variance smoothing	Unscaled	Quadratic (from Gaussians)
SVM (RBF)	C = 1.0, gamma = scale, probability = True	Scaled	Smooth non-linear, max margin
ANN (MLP)	64–32 ReLU, Adam, early stopping	Scaled	Flexible non-linear

Part 3 – Member 3: Binu Regmi (S392323)

Overview of Model Evaluation Measures, Results Obtained, and Overall Understanding • *1,463 words*

5. Overview of Model Evaluation Measures

5.1 The Confusion Matrix

We evaluated all the metrics we used using the **confusion matrix**. The confusion matrix counts the four possible outcomes when making a binary prediction. The four outcomes include: **True Positive (TP)**, a positive case correctly predicted as positive; True Negative (TN), a negative case correctly predicted as negative; False Positive (FP), a negative case incorrectly predicted as positive; and **False Negative (FN)**, a positive case incorrectly predicted as negative. One important aspect of how to count "which class" is whether it is considered "positive". In scikit-learn's representation of our data set, "benign" = 1 and "malignant" = 0. By default, scikit-learn uses label 1 as the positive class for the precision and recall functions.

Thus, our precision, recall, and F1 values presented in Table 4 are based upon the **benign class**.

However, because missing a malignant tumour would result in the greatest clinical harm, we have also reported the malignant class values in Table 5.

5.2 Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Accuracy is the proportion of all predictions that are correct. It is easy to understand and is a good overall summary when classes are reasonably balanced. Its weakness is that it treats all errors equally and can hide poor performance on the minority class. With 62.7% benign samples, a useless model that always predicts “benign” would still achieve 62.7% accuracy.

5.3 Precision

Precision = TP / (TP + FP)

Precision answers the question: *of all the samples the model labeled as a given class, how many really belong to it?* High precision means few false alarms. For the malignant class, high precision means few healthy patients are wrongly told they may have cancer and sent for unnecessary further procedures.

5.4 Recall (Sensitivity)

Recall = TP / (TP + FN)

Recall answers the question: *of all the samples that really belong to a class, how many did the model find?* In medicine, recall of the disease class is called **sensitivity**. For breast cancer diagnosis, **malignant recall is the most important measure**, because every malignant case that is missed is a patient whose treatment could be delayed. There is usually a trade-off

between precision and recall: making a model more cautious (flagging more samples as malignant) raises recall but lowers precision.

5.5 Supporting Measures: F1-Score and ROC-AUC

F1 = 2 × (Precision × Recall) / (Precision + Recall)

The **F1-score** is the harmonic mean of precision and recall. It is only high when both are high, so it is a useful single number for balancing the two. The **ROC-AUC** measures something different: rather than judging the final labels at a single threshold, it uses predicted probabilities and assesses how well the model ranks positive cases above negative ones across all possible thresholds (Fawcett, 2006). An AUC of 1.0 means perfect ranking, and 0.5 means random guessing. A model with a high AUC can be tuned to favor recall simply by lowering its decision threshold.

6. Results Obtained

6.1 Overall Performance

Table 4 shows the performance of the five models on the 114-sample stratified test set. These are exactly the values produced by our code (*model_results.csv*).

Table 4: Test-set results (precision, recall, and F1 for the benign class, scikit-learn default)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	0.9211	0.9565	0.9167	0.9362	0.9163
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9932
Naive Bayes	0.9386	0.9452	0.9583	0.9517	0.9878
SVM (RBF)	**0.9825**	**0.9861**	**0.9861**	**0.9861**	**0.9950**
ANN (MLP)	0.9474	0.9459	0.9722	0.9589	0.9927

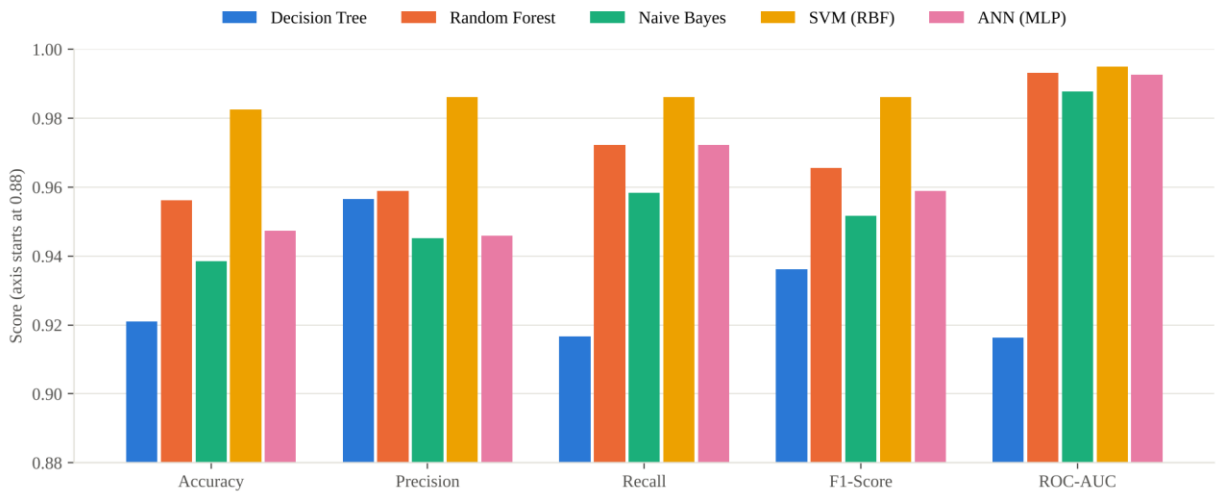


Figure 13: Comparison of all five metrics across the models (y-axis starts at 0.88)

What we learned: All five models exceed 92% accuracy, far above the 62.7% "always benign" baseline. This confirms what the PCA plot suggested: the classes are largely separable. The **SVM (RBF)** is the best model across all metrics, achieving 98.25% accuracy (112 of 114 correct). Random Forest is second (95.61%, 109 correct), followed by the ANN (94.74%, 108), Naive Bayes (93.86%, 107), and the Decision Tree (92.11%, 105). The chart also shows a split in ROC-AUC: four models score above 0.98, while the single Decision Tree is clearly lower at 0.916. Note that the y-axis starts at 0.88 to make the differences visible; in absolute terms, all models are close.

6.2 Confusion Matrices and the Malignant Class

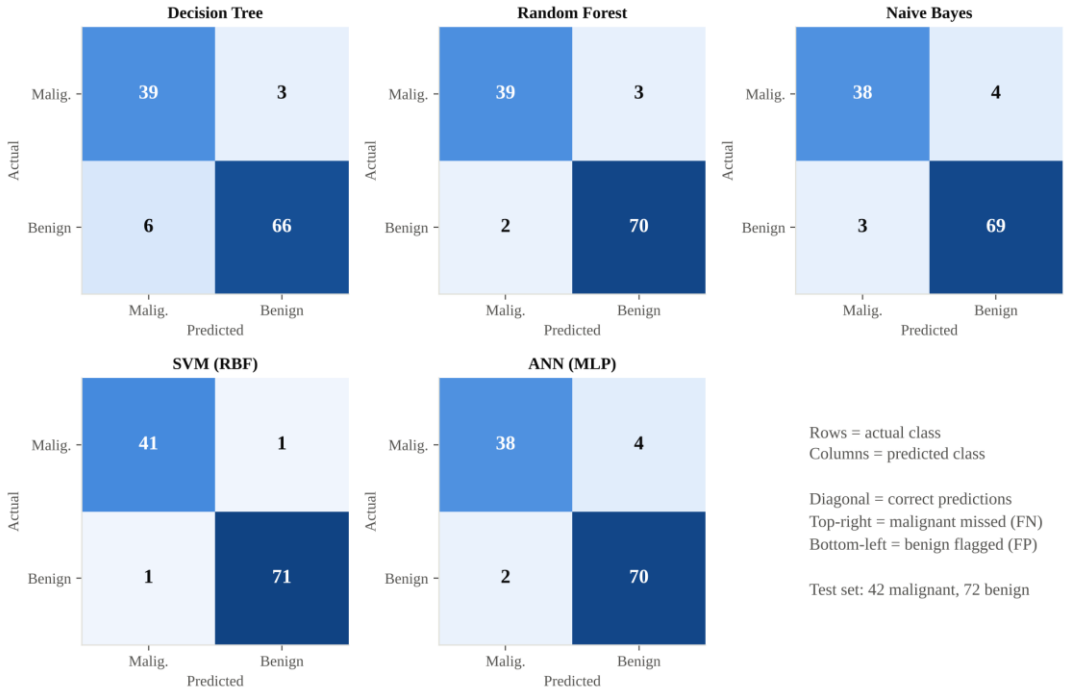


Figure 14: Confusion matrices on the test set (rows = actual, columns = predicted)

Table 5: Errors and metrics with malignant as the positive class (42 malignant test cases)

Model	Malignant missed (FN)	Benign flagged as malignant (FP)	Malignant precision	Malignant recall
Decision Tree	3	6	0.8667	0.9286
Random Forest	3	2	0.9512	0.9286
Naive Bayes	4	3	0.9268	0.9048
SVM (RBF)	**1**	**1**	**0.9762**	**0.9762**
ANN (MLP)	4	2	0.9500	0.9048

What we learned: When you look at the results on the malignant side, the image changes slightly. The SVM missed just one of 42 malignant tumours (malignant recall 97.6%) and made only one false alarm. Both the random forest and the decision tree each missed three malignant cases; however, Naive Bayes and the ANN each missed four, resulting in a malignant recall of

only 90.5%.Although the ANN had the same benign recall as the random forest (0.9722), it also missed one additional cancer case. False alarms are where the decision tree is weakest. It falsely identified six benign tumours as malignant.

Therefore, the malignant precision of the decision tree was lowest (86.7%). These examples show how looking at overall accuracy, or only at the default positive class, can hide important differences between models.

6.3 ROC Curves

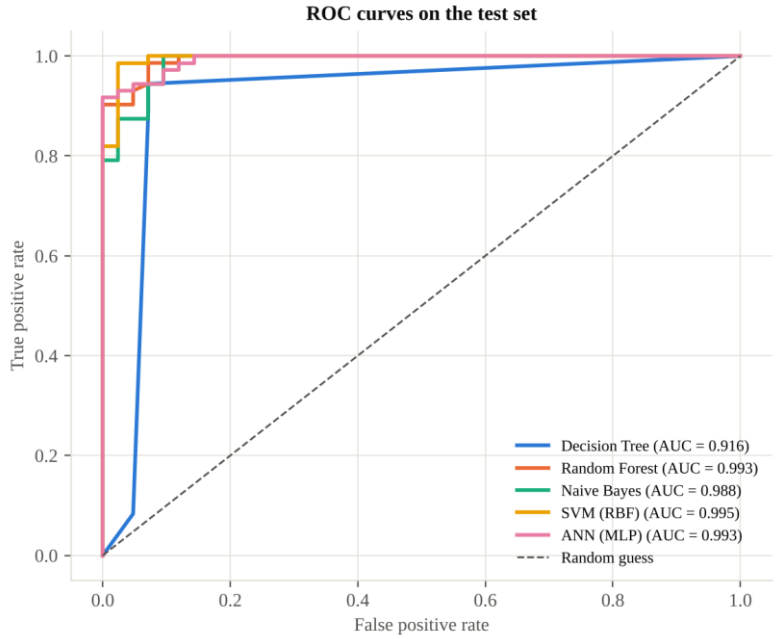


Figure 15: ROC curves for the five models on the test set

What we have discovered about the performance of SVM, Random Forest, ANN, and Naive Bayes is that they produce very similar curves. Each curve begins to go up rapidly and reaches a true positive rate of 0.9 before the false positive rate reaches zero.

Therefore, each method produced a Probability Output ranking nearly all benign samples above all malignant ones. ANN's AUC (0.993), like Random Forest's, was among the highest of the methods tested.However, ANN had a lower Accuracy than Random Forest. ANN produced errors due to the use of a default threshold value of 0.5. As such, ANN could be made better through Threshold Tuning. Only a small number of Corner Points were visible on the Decision Tree Curve. Because a Decision Tree with five levels of depth produces only a few distinct Probability Values (one for each Leaf), its low Resolution accounts for its much lower AUC of 0.916

6.4 Feature Importance

```
importances = pd.Series(models["Random Forest"].feature_importances_,
                        index=X.columns).sort_values(ascending=False)
print(importances.head(10).round(4))
```

Figure 16:Code: extracting feature importance from the Random Forest

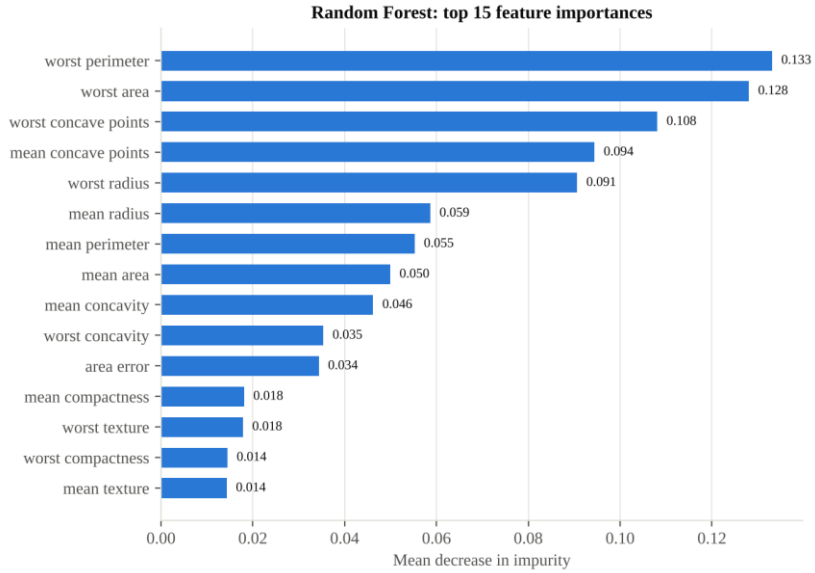


Figure 17: Random Forest: top 15 features by mean decrease in impurity

What we learned: The three most important variables for predicting malignancy are worst perimeter (0.133), worst area (0.128), and worst concave points (0.108), followed by mean concave points and worst radius. These five variables account for 55% of the overall importance. By the numbers, the "worst" variables contributed 56% to overall importance, the "mean" variables contributed 35%, and the standard error variables contributed only 9%. This supports the initial exploratory analysis that the size of the largest nucleus and the irregularity of its boundary (concavity) are the best predictors of whether it is malignant or benign. Clinical experience has also shown that malignant cells tend to have a larger size and/or irregular shape compared to benign cells (Street et al., 1993). Variables identified as being of little use in Section 3, such as fractal dimensions and symmetry, were not among the top-ranked variables.

7. Overall Understanding and Conclusion

7.1 What the Results Tell Us

The SVM was the top performer in our study because many of the assumptions made about this dataset were met. The data used for modelling is relatively small, numerical, and has been standardized prior to analysis. The two classes are almost completely separated, with a curvilinear decision boundary apparent in the pair plots of the variables. The maximum-margin decision boundary depends only on 97 of the observations (support vectors) and generalizes effectively from very few examples. The random forest was the second highest performing model, as it averages over 200 decision trees to minimize the variability inherent in any single

tree. The artificial neural network did not improve over traditional machine learning approaches. These findings align with previous studies that have demonstrated that, on small to medium-sized tabular datasets, deep learning typically fails to surpass appropriately selected traditional models (Grinsztajn et al., 2022).

While neural networks perform extremely well when learning features directly from large quantities of raw, unstructured data (e.g., images), the features employed in this study were engineered by experts.

Therefore, there were few opportunities for the intermediate layers of the network to identify additional features and the limited training set limited the ability to accurately estimate the 4097 parameters associated with the network.

7.2 Limitations

- Single Train/Test Split – Since we have only 114 test samples, each correct prediction will equal 0.88% of our accuracy. Therefore, the difference in accuracy between the Random Forest (109 correct) and Artificial Neural Network (108 correct) is only one sample. If we were to use a different random split for the train/test sets, it would be possible to see a different ranking among the three middle-performing models.

- Default Positive Class – As stated previously, all results presented in our primary table follow the Scikit-learn convention of using "benign" as the positive class. The results for the malignant-class metrics, which are more relevant to clinical practice, are provided separately in Table 5.

- No Hyperparameter Tuning – We did not tune any hyperparameters for any of the models; therefore, the results obtained with the default or simple parameter settings may differ greatly once the models' hyperparameters have been optimized.

- One Data Source – All images used to create the dataset came from a single institution. As such, we cannot determine how well the models would perform on images created at different institutions or using different imaging equipment.

7.3 Future Work

Future studies will need to implement stratified k-fold cross-validation to get accurate averages and confidence intervals for all of the models, and GridSearchCV to find optimal values for hyperparameters, including C and gamma for the SVM model, number of trees and depth for the Random Forest model, and layer size and learning rate for the ANN model.

Additionally, lowering the decision threshold to prioritize malignant recall could help improve accuracy. Finally, using a larger image-based dataset will allow researchers to see if deep

learning models (like convolutional neural networks) have an advantage over traditional models in the ability to learn from raw image data.

7.4 Conclusion

The results of this study show that machine learning can determine whether a woman has a benign or malignant tumour from her fine-needle aspirate (FNA) images, based on measurements taken from those images, and can be very effective. The results show that all five models used in the study achieved accuracies greater than 92%; the SVM model using an RBF kernel was the most successful, achieving 98.25% accuracy, a ROC-AUC of 0.995, and missing only one malignant case out of 42. The feature importance analysis and the exploratory charts both indicate that tumour size and border irregularity are the two major factors in determining whether the tumour is malignant. With additional validation studies, these models may be able to offer a quick and reliable second opinion to assist clinicians with their decision-making process.

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